

1 Reliable comparison of severe felt-intensity
2 classifiers with versioned and dependent event
3 data: a USGS DYFI benchmark

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9 **Abstract**

10 Engineers comparing machine-learning classifiers on evolving event catalogues
11 can reach false conclusions when dependent events cross data splits or small score
12 differences are treated as wins. We ask whether candidate severe felt-intensity
13 classifiers comply with predefined rules for allowed inputs and evaluation, and
14 whether their apparent improvements remain supported after catalogue version,
15 event grouping, temporal evaluation, and paired uncertainty are fixed. Using a
16 frozen USGS “Did You Feel It?” (DYFI) snapshot of 7,520 $M \geq 5$ records, we
17 formed 2,362 eligible events, reserved 430 events from 2023–2024 as a tempo-
18 ral holdout, and evaluated fixed baselines using Brier score and sequence-cluster
19 bootstrap intervals. All four learned models improved on the no-skill reference;
20 paired uncertainty did not distinguish the two lowest-Brier models, and the input
21 rules excluded a candidate using event year. For a fixed random-forest diagnos-
22 tic, the observed differences between sequence-grouped and random folds were
23 small (Brier -0.006 ; AUROC $+0.012$); this comparison is descriptive because
24 no interval was retained. The benchmark helps engineers reject inadmissible
25 comparisons and unsupported winner claims. It is intended for transparent retro-
26 spective comparison; confirmatory or operational use would require fresh hidden
27 data and separate validation.

28 **Keywords:** model comparison, evaluation reliability, benchmark qualification,
29 dependent-event data, reproducibility, USGS DYFI

1 Introduction

Engineering model selection depends on comparisons that remain valid when a dataset is updated, observations are dependent, and leading scores are close. Without those controls, an apparent improvement can be caused by a changed population, a proxy for the outcome or evaluation axis, cross-split dependence, or unquantified sampling variation rather than by a better model. Standard benchmarking is valuable precisely because it fixes a common task and metrics; recent work in this journal, for example, compares object detectors on COCO to support model selection for engineering applications [29]. The reliability problem is harder for event catalogues because both the catalogue and the dependence structure can evolve.

The U.S. Geological Survey “Did You Feel It?” (DYFI) system is a useful engineering case study. It converts crowdsourced felt reports into a community-decimal intensity (CDI) for each earthquake and is widely reused as an interpreted data product [2, 22, 34, 35]. The target studied here is retrospective classification of whether event-maximum CDI reaches 6 from outcome-blind source metadata. It is not a real-time alert, causal model, or operational triage rule.

Our engineering question is: *do performance and model-superiority claims remain supportable when catalogue version, dependent-event splitting, temporal evaluation, paired uncertainty, and reconstruction are controlled?* The intended users are applied machine-learning engineers evaluating a candidate classifier, maintainers publishing a stable benchmark, and reviewers auditing a claimed improvement. They need two decisions: whether a candidate is admissible for comparison and whether its improvement over a declared reference is supported rather than a point-score fluctuation. Existing felt-intensity studies use random or stratified partitions and do not provide that reusable decision contract [15, 20].

1.1 Contributions

We address unreliable model comparison by releasing a versioned benchmark and an executable candidate-qualification contract. Concretely, the work provides:

1. **A single-source, version-frozen DYFI benchmark** for the task “outcome-blind source metadata $\rightarrow$ later event-maximum $\max\text{CDI} \geq 6$ ”, frozen to one dated access instant so that any later catalogue revision defines a new version rather than a silent overwrite (Section 3).
2. **An executable six-category leakage audit** that rejects direct-label, post-outcome-ascertainment, identifier-proxy, and downstream-product features, and rejects cross-role sequence straddles and temporal-cutoff violations, accepting *only* the exact source-only feature set; it is exercised by 110 unit tests and rejects every deliberately contaminated test case (Section 4.1).
3. **Internally specified, outcome-independent roles** built on order-stable space-time sequence grouping and a temporal external holdout, so that split membership never uses the label (Section 4.2).
4. **A fixed, non-selective baseline suite** with calibration-aware proper scoring, Brier-score sequence-cluster bootstrap uncertainty, explicit handling of ties and unavailable calibration estimates, and single-use authorization of the decision-bearing protected evaluation (Sections 4.4–4.5).
5. **A candidate-qualification workflow** that fails closed on forbidden features or invalid roles and supports an improvement only when paired Brier-score uncertainty excludes zero (Section 5.5).
6. **A quantified split-sensitivity diagnostic** (naive random versus sequence-grouped splitting), together with a reproducible release bundle and machine-readable verification record (Section 5).

The learned models are worked candidates, not proposed production systems. Their comparison demonstrates the decisions the contract permits: all admissible

Table 1 Engineering failure modes addressed by the benchmark qualification contract

Failure source	Invalid engineering conclusion	Qualification control
Catalogue revision	Non-identical event population	Frozen access instant + SHA-256 release
Outcome-derived field	Label leakage	F1/F2 schema audit
Identifier or downstream proxy	Proxy leakage	F3/F4 field audit
Sequence crosses roles	Dependent-event contamination	F5 sequence-role audit
Temporal-cutoff violation	Future information in development	F6 temporal audit
Point-score ranking	False model-superiority claim	Paired sequence-cluster interval
Environment drift	Irreproducible scores/artifacts	Pinned environment + reconstruction hash

82 learned candidates improve over no-skill, one candidate is outside the declared
83 source-only schema, and the two lowest-Brier models remain unresolved.

84 1.2 Engineering significance and intended use

85 The output is an engineering test harness for transparent retrospective model compar-
86 ison. A declared candidate enters through the released schema, is excluded if it uses
87 forbidden outcome-derived fields or violates role isolation, is scored on the fixed tempo-
88 ral holdout, and is compared with paired sequence-cluster uncertainty. This workflow
89 prevents an engineer or reviewer from declaring superiority from a different catalogue
90 version, a contaminated split, or an unresolved point difference. Because holdout labels
91 are public, the release is not an untouched prospective test: confirmatory evaluation of
92 a new model requires a fresh benchmark version or hidden evaluation service. Table 1
93 names the failure prevented by each control, and Fig. 1 shows the decision path.

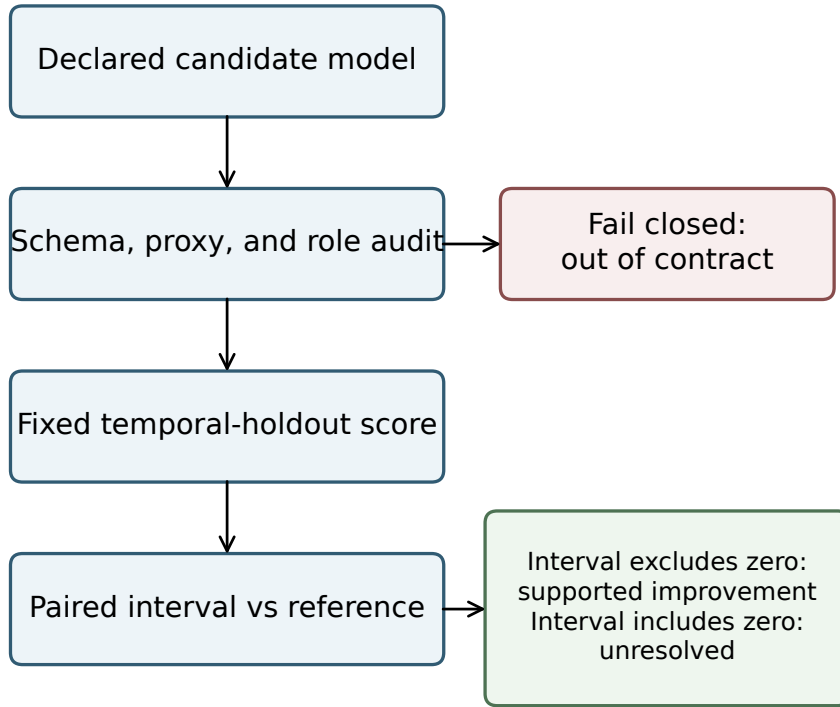


Fig. 1 Candidate-model qualification. A candidate that fails the schema, proxy, role, or temporal audit is inadmissible. An admitted candidate is scored on the fixed holdout and compared with paired uncertainty; an interval including zero remains unresolved rather than becoming a winner claim

2 Related work and novelty

DYFI and macroseismic intensity.

The DYFI system and its CDI construction, outlier handling, and post-event revision behaviour are documented in the foundational and review literature [2, 22, 34, 35]; these establish that CDI is an after-the-event, response-driven quantity subject to ascertainment effects, which is precisely why our protocol uses response count only to determine eligibility and forbids it as a predictive feature (Section 3). Rösler et al. [24] further show specifically for DYFI that interpreted products can change and should

102 be preserved with timestamps for reproducible analysis. The dated, hashed snapshot
103 released here implements that preservation principle. We reuse these provenance facts
104 rather than re-deriving them.

105 *Domain benchmark resources.*

106 The INGe resource links Italian macroseismic intensities to colocated strong-motion
107 records for benchmarking intensity–ground-motion relationships [18]. It is a domain-
108 near resource precedent, but differs from the present event-level USGS DYFI
109 classification task, source-only feature boundary, leakage audit, and protected tem-
110 poral evaluation. The 2026 USGS Western North America resource contains about
111 696,000 site-level macroseismic observations from about 5,600 earthquakes and is pre-
112 dominantly DYFI [6]. It is a larger observational resource, but does not define this
113 event-maximum binary task, executable leakage audit, or frozen temporal evaluation.
114 Stockman et al. [28] provide the nearest broader leakage-aware seismological bench-
115 mark precedent: EarthquakeNPP standardizes public catalogues, repairs an improper
116 earthquake-forecasting split, compares neural point processes with ETAS, and finds
117 no consistent neural-model advantage. Its target is occurrence forecasting with chrono-
118 logical likelihood/CSEP evaluation, rather than source metadata predicting a later
119 citizen-reported felt-intensity endpoint; the resources are therefore methodologically
120 adjacent but task-distinct.

121 *Engineering benchmark and model-comparison precedents.*

122 Comparative engineering studies use a fixed task and common metrics to turn model
123 outputs into selection evidence. Tian et al. [29], for example, compare object detec-
124 tors on COCO to support model choice for smart-city applications. In this journal,
125 machine-learning comparisons have also been tied to explicit engineering decisions in
126 rock-property estimation and pavement maintenance [1, 17], while seismic engineering

127 studies state the response quantity and design interpretation affected by the analy-
128 sis [26]. The present contribution follows that decision-oriented form but evaluates
129 the reliability of the comparison itself: whether a candidate is admissible, whether it
130 improves over a reference, and whether the evidence supports naming a winner.

131 An adjacent event-level binary study used post-event LastQuake felt reports to
132 classify high- versus low-impact earthquakes and evaluated later 2022 events [14]. Its
133 crowdsourced predictors, impact endpoint, and rapid-response objective differ from
134 the source-only DYFI benchmark and reusable evaluation contract developed here.

135 *Macroseismic databases and citizen observations.*

136 Event-specific analysis in the 2019 Athens earthquake combined online felt reports
137 with ground motion and source modelling [12]. Crowdsourced macroseismic observa-
138 tions have also been used to characterize geological settings and spatial variations in
139 shaking response [25]. Curated macroseismic resources such as the Belgian traditional
140 macroseismic database and MACROSISDATA preserve historical intensity observa-
141 tions for transparent reuse and later analysis [16, 27]. The present work extends this
142 seismological resource tradition from data preservation to a versioned, executable
143 contract for model evaluation.

144 *Machine learning on felt/macroseismic intensity.*

145 Recent studies predict macroseismic intensity with random forests and gradient boost-
146 ing from instrumental or catalogue features [15, 20]. Machine learning has also been
147 applied to earthquake alert-level classification using intensity-related features and to
148 waveform detection with openly released seismic datasets [5, 19]. Those tasks use post-
149 event intensity descriptors or waveforms, rather than the source-only feature boundary
150 studied here. Each felt-intensity model application is evaluated under a random or
151 stratified partition and does not release a reusable, event-grouped, leakage-audited
152 benchmark; Mori and Naso [15] moreover predict from instrumental ground-motion

parameters rather than outcome-blind catalogue metadata, and Patelli et al. [20] evaluate a single-system train/test split on seismometric inputs. Their concrete drawback for our purpose is the evaluation contract: a random or stratified split does not remove within-sequence near-duplicate contamination, and none ships an executable leakage audit or a version-frozen resource that a third party can rerun. Our protocol addresses that specific drawback with internally specified sequence-grouped roles and an executable six-category audit.

DYFI-based intensity models and concurrent leakage-aware work.

Closest to our exact data source, Karthik Kumar et al. [10] predict DYFI community intensity from the same catalogue metadata (magnitude, depth, location) with a random-forest regressor on Indian events under a single naive train/test split (reported $R^2 = 0.829$); our resource does not fit a competing regressor but instead supplies a versioned, digitally fingerprinted, event- and sequence-grouped benchmark with an executable leakage audit and a protected decision-bearing evaluation authorized once, in place of the naive split that setup uses. Concurrently, Li et al. [13] release a DYFI-grounded benchmark that simulates Modified Mercalli Intensity (MMI) for two earthquakes (2014 Napa and 2019 Ridgecrest) with large language models and includes an explicit location-identifier leakage test; our resource differs in task and construction—an outcome-blind source-metadata→later-severe binary classification over a decade-scale $M \geq 5$ corpus with internally specified sequence-grouped roles, rather than multimodal per-zip-code MMI simulation for two events—so the two are complementary rather than duplicative.

Cross-validation for dependent data.

That grouped or blocked cross-validation is the correct evaluation strategy for temporally, spatially, or hierarchically structured data is established practice [11, 23]; we do not claim this as novel.

179 ***Novelty delta.***

180 Conceding the above, the delta this resource supplies that prior work does not is the
181 *released, versioned, hashed, single-source DYFI benchmark* itself: an executable six-
182 category audit covering forbidden fields, proxy variables, and split contamination; an
183 internally specified, sequence-grouped role assignment with a temporal external hold-
184 out, a documented severe endpoint with threshold and threshold sensitivity, a fixed
185 non-selective baseline suite with a no-skill comparator, a calibration/uncertainty/tie
186 contract, an exact-reproduction verification, and a quantified naive-versus-grouped
187 split diagnostic. A current-year search located no released resource occupying this
188 exact combination; novelty is stated as a concrete delta to located work, with no
189 first/only superlative asserted.

190 **3 Data, provenance, and outcome-boundary** 191 **disclosure**

192 ***Source and single-instant pinning.***

193 The corpus is derived entirely from USGS DYFI and its parent ANSS Comprehensive
194 Catalog (ComCat), retrieved through the International Federation of Digital Seis-
195 mograph Networks (FDSN) event web service [30, 33]. The acquisition envelope is
196 outcome-independent: $M \geq 5$, product type `dyfi`, event type `earthquake`, starttime
197 2015-01-01 to endtime 2025-01-01. FDSN defines the end bound as inclusive; no frozen
198 record occurs at that boundary, so the realized snapshot spans 2015–2024. Results are
199 ordered by time. The preliminary count request and the final data request use this
200 same envelope. Because the catalogue evolves, the downloaded data were frozen at
201 one access instant (2026-09-01T06:33:30Z) and assigned a SHA-256 digital fingerprint.
202 A preliminary count changed from 7,575 to 7,576 during the acquisition session, doc-
203 umenting one live-catalogue revision. The final downloaded snapshot contained 7,520
204 DYFI records, and a count requested 50 seconds later in the same acquisition session

also returned 7,520. Thus, the one-record change belongs only to the earlier preliminary checks; it is not a discrepancy in the frozen analysis snapshot. Likewise, the 56-record difference between the last preliminary count and the final 7,520-record snapshot is not an analysis exclusion: the preliminary count had no paired payload, so its membership difference cannot be decomposed and is retained only as catalogue-version provenance. The near-contemporaneous final query and count agree exactly. The digital fingerprint of the downloaded bytes, rather than a preliminary count, defines this resource version. The frozen snapshot contains 7,520 DYFI records (Table 2, Fig. 2).

Protocol chronology.

The file records an internal 2026-09-01 freeze but was first publicly timestamped on 2026-09-19. We therefore describe it as an internally frozen protocol, not an externally registered preregistration.

Retrospective eligibility and post-report ascertainment.

DYFI intensity is an after-the-event quantity whose reliability grows with the number of felt reports [22]. To keep the label meaningful we apply a data-informed eligibility filter of at least ten responses per event. The response count is used *only* for eligibility and is explicitly forbidden as a predictive feature, because it is a post-outcome ascertainment proxy; this forbiddance is enforced by the leakage audit (family F2, Section 4.1). Eligibility yields 2,362 events with severe prevalence 0.298 (703 severe events). Of these, 2,360 events are assigned to the internally specified analysis roles (Section 4.2) and contain 702 severe events; the remaining two are temporal-straddle exclusions, one of which is severe. That single severe straddle event accounts for the difference between the eligible cohort (703 severe) and the analysis-assigned set (702 severe), so eligibility is not the same as analysis-role assignment (Table 2).

229 ***Endpoint.***

230 We label an event severe when its event-maximum CDI is at least 6.0 (intensity VI or
231 greater). We use 6.0 as the primary threshold and report sensitivity at 5.5, 6.0, and
232 6.5. We also examine a near-threshold band of ± 0.5 (Section 5.2).

233 ***Outcome-boundary disclosure and held-out semantics.***

234 Only corpus-level outcome-blind counts and inherited aggregate feasibility statistics
235 were exposed before construction; no per-event label informed the endpoint, split, or
236 tuning choices, and role membership is assigned by an internally specified, outcome-
237 independent rule that never uses the label. We therefore describe the external holdout
238 as *held out from model development with aggregate feasibility previously known*, and we
239 deliberately do *not* call it “untouched” or “confirmatory.” Two event identifiers were
240 prelisted for quarantine because their aggregate counts had been seen during source
241 verification; neither identifier occurs in the frozen snapshot, so the rule is non-operative
242 in version 1 and excludes no record. The complete list of protocol decisions is reported
243 in Section 5.8. No respondent-level personally identifying information is redistributed;
244 only event-level source metadata, the severe label, the eligibility-only response count,
245 identifiers, and role assignments are released. Snapshot-level provenance is supplied
246 by the activated source-locator manifest and source payload SHA-256; version 1 does
247 not claim per-event product-version hashes. The release bundle follows datasheet-
248 for-datasets documentation [8] and findable, accessible, interoperable, and reusable
249 (FAIR) data principles [36].

250 4 Benchmark design

251 4.1 Feature sets and the six-category leakage audit

252 The primary *source-only* feature set contains magnitude and depth. It is the only
253 feature set evaluated on the temporal holdout. An *expanded-metadata* sensitivity anal-
254 ysis (baseline B3) additionally uses latitude, longitude, and event year. Event year is
255 known at prediction time and is not intrinsically label leakage, but it is excluded by the
256 declared source-only temporal-holdout contract. The audit accepts only an approved
257 feature set with the expected fields in the expected order; any extra or unknown field
258 is rejected.

259 The executable audit enforces six leakage families. Field-level: F1 direct label or
260 its constituents; F2 post-outcome ascertainment or revision (including the response
261 count, eligibility-only); F3 identifier or source proxy; F4 downstream-product proxy.
262 Split-level: F5 sequence cross-role straddle; F6 temporal-cutoff violation. An axis-
263 variable check additionally rejects origin year under a temporal holdout and, when a
264 geographic holdout is configured, latitude/longitude/region. The latter audit branch
265 is provided for reuse but was not executed as an external evaluation in this study.
266 This enforces a conservative B3 scope restriction; it is not evidence that event year
267 is a leaking label proxy. The audit is useful independently of model performance: it
268 must reject every deliberately contaminated test case and accept only the approved
269 clean feature set. It is exercised by 110 unit tests across four suites, all passing, and
270 accepted the source-only feature set in the frozen snapshot.

271 4.2 Sequence grouping and outcome-independent roles

272 Events are grouped into seismic *sequences* by order-stable single-link space-time
273 connected components (100 km, 30 days); the sequence, not the event, is the unit
274 of role assignment, so events linked by this fixed rule cannot straddle roles. This
275 rule is a deterministic grouping heuristic, not a validated catalogue of all physical

mainshock–aftershock relationships; unlinked true aftershocks may therefore remain in different roles. Roles are assigned by an internally specified, outcome-independent rule (fixed assignment key, temporal cutoff year 2023, development fraction 0.2, five cross-validation folds); the rule provably never reads the label. The external holdout contains events on or after 2023. The frozen internal protocol also named a secondary leave-region-out geographic evaluation, but that branch was not executed and no geographic role manifest is claimed. The reported geographic result instead summarizes the temporal-holdout predictions by 10° cell (Table 9); Table 8 records this deviation. Digital fingerprints of the completed temporal role assignments are published so that third parties can verify them. The partition is exhaustive and disjoint: $1,573 + 357 + 430 + 2 = 2,362$ events, comprising 1,573 train, 357 development, 430 external temporal holdout, and 2 temporal-straddle exclusions (Table 2, Fig. 2). Construction (train plus development) is 1,930 events across 1,530 sequences; the holdout is 430 events across 354 sequences.

4.3 Baseline suite and exclusions

The fixed suite is B0 no-skill (constant prevalence), B1 magnitude-only logistic regression, B2 source-only logistic regression, B3 expanded-metadata logistic regression (used only in compatible sensitivity analyses), B4 random forest (scikit-learn; 21), and B5 XGBoost [4]. Each learned baseline is tuned *only* by the protocol-defined deterministic grouped cross-validation on the construction corpus over a fixed grid, named before evaluation, and carried unchanged to the protected holdout with frozen best parameters. A machine-readable fit log enforces a maximum of 120 model fits; the executed run used 91. Two comparators are excluded for protocol-defined, reproducible reasons recorded in the protocol-decision table (Section 5.8): B3 is not evaluated on the temporal holdout because origin year is outside the declared source-only feature contract; and the Atkinson–Wald (2007) DYFI intensity-prediction equation [2]

is excluded because it cannot be applied to the source-only feature set and tectonically heterogeneous population without inventing a site-distance covariate, a per-event tectonic-regime assignment, and a probability link.

4.4 Metrics, calibration, uncertainty, and ranking

The primary score is the Brier score [3]; secondary scores are log-loss, AUROC, and calibration intercept/slope [31]. For numerical stability, log-loss clips predicted probabilities symmetrically to $[10^{-15}, 1 - 10^{-15}]$ before taking logarithms. Brier-score uncertainty is quantified by a sequence-cluster bootstrap ($n_{\text{boot}} = 2000$, 95% intervals) that resamples whole sequences rather than events [7], with a minimum-valid-replicate rule. Calibration and feasibility states are *derived*, not hard-coded: a constant-prevalence predictor has no identifiable calibration slope, so the table reports that value as unavailable. Pairwise ranking uses a paired sequence-cluster bootstrap Brier difference with an explicit tie probability; the table reports whether each 95% interval includes zero, together with a tie-aware joint ordering (Section 5.6). Because the intervals use 2,000 Monte Carlo draws, near-zero endpoints are interpreted cautiously rather than as stable ranking guarantees. The protocol requested intervals for every metric; only Brier-based intervals were retained, so other metrics are descriptive point estimates (Table 8).

4.5 Protected evaluation and reproducibility

The official decision-bearing protected evaluation was authorized once by an automated, globally consumed token; the one-shot wrapper refuses a second authorized run. After the result was fixed, the token-free deterministic analytic core—including holdout scoring—was replayed twice for verification, and both replays reproduced the recorded result fingerprint exactly. Thus, single-use refers to result authorization and decision-bearing use, not to the total number of post-result computations. The frozen holdout predictions were also recovered bit-for-bit from the released rows and frozen

best parameters (reported in Section 5.8), under a pinned scientific-Python environment (NumPy, SciPy, scikit-learn, XGBoost; 4, 9, 21, 32). All fits, exclusions, subsets, cutoffs, and pairwise comparisons are reported regardless of outcome; the only permitted omissions are the protocol-defined exclusions above. Computation is local-only under a pinned environment.

Reproducible artwork and generative AI assistance.

All figures were generated with Matplotlib 3.11.1 directly from the frozen analysis assets; no manual image manipulation or AI-generated artwork was used. ChatGPT assisted language editing, formatting, code review, literature discovery, critical consistency review, and revision suggestions. It did not generate data, execute model fits, or create the reported numerical results. The authors independently verified every cited source, computation, interpretation, and assisted change; all scientific judgments and final decisions remained with the authors, who retain full responsibility for the manuscript and released artifacts.

5 Results

All result-bearing values below are generated from the frozen analysis outputs and checked automatically against their source records; no result cell was hand-edited.

5.1 Cohort and roles

Table 2 and Fig. 2 give the cohort accounting. Of 7,520 frozen DYFI records, 2,362 events pass the ten-response eligibility filter (severe prevalence 0.298; 703 severe events), partitioned into 1,573 train, 357 development, and a 430-event temporal hold-out, with two temporal-straddle exclusions. Thus 2,360 of the 2,362 eligible events receive analysis roles; they contain 702 severe events, and the 703rd severe event is one of the two straddle exclusions. Two identifiers prelisted for quarantine are absent

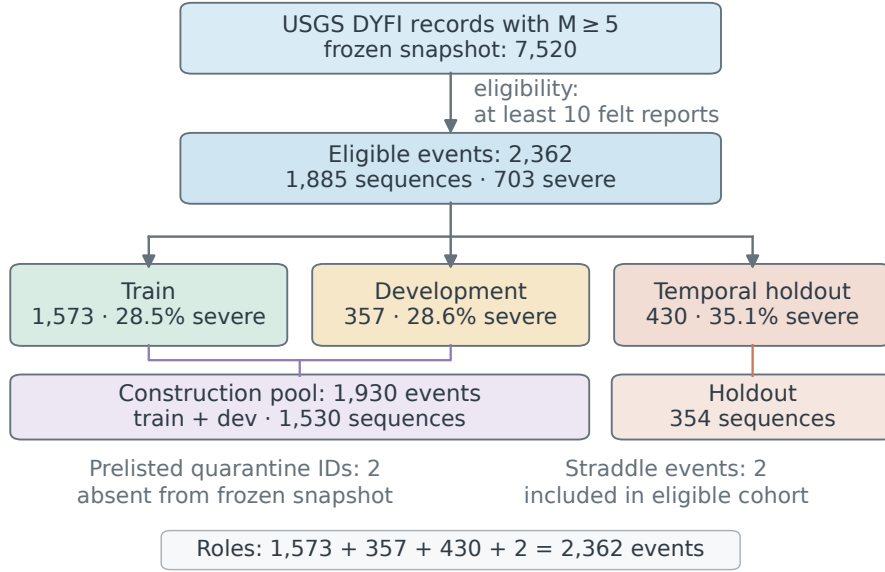


Fig. 2 Cohort construction and role assignment. On the frozen USGS DYFI $M \geq 5$ snapshot. The 7,520 source records yield 2,362 eligible events in 1,885 seismic sequences. Of these, 2,360 are assigned to train ($n = 1,573$), development ($n = 357$), or temporal holdout ($n = 430$); two straddle events are excluded from role assignment

Table 2 Cohort accounting and outcome-independent role assignment

Stage	Events	Severe	Severe rate	Sequences
USGS DYFI records with $M \geq 5$ (frozen snapshot)	7520	–	–	–
Eligible (at least 10 felt reports)	2362	703	0.298	1885
Train	1573	449	0.285	–
Development	357	102	0.286	–
Construction (train+dev)	1930	551	0.285	1530
External temporal holdout	430	151	0.351	354
Temporal-straddle excluded	2	1	–	1

Note. Roles were fixed without using outcomes. Two event identifiers prelisted for quarantine are absent from the frozen snapshot, so the safeguard removes no version 1 record. Severe = 1 when an event's maximum community decimal intensity (CDI) is at least 6. The eligible severe count reconciles as 703 = 449 (train) + 102 (development) + 151 (holdout) + 1 (straddle exclusion). Thus the 2360 role-assigned events contain 702 severe events.

352 from the frozen snapshot, so that safeguard removes no version 1 record. Construc-
 353 tion severe prevalence is 0.285 and holdout prevalence is 0.351; the higher holdout
 354 prevalence reflects the temporal transport shift and is reported, not adjusted.

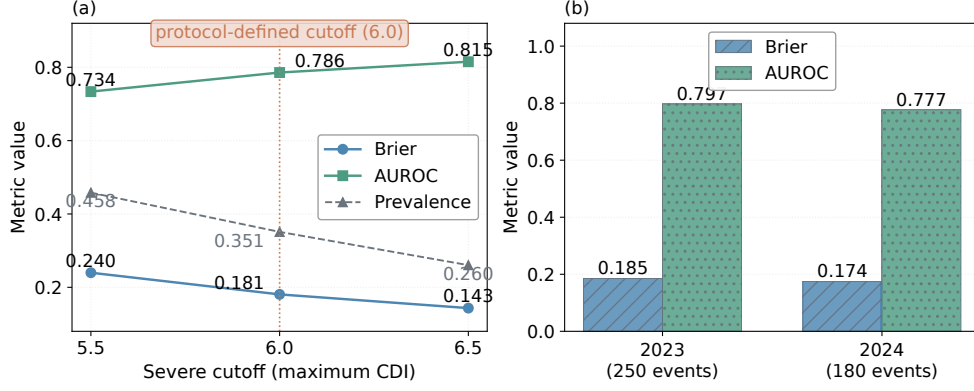


Fig. 3 Endpoint and temporal sensitivity. Results for the source-only logistic regression (B2). (a) Performance as the severe-intensity cutoff varies. (b) Performance within the 2023 ($n = 250$) and 2024 ($n = 180$) portions of the temporal holdout. The cutoff analysis rescores fixed B2 probabilities without refitting. Lower Brier score and higher AUROC are better

Table 3 Endpoint-threshold probe using fixed source-only probabilities

Maximum CDI cutoff	Severe rate	Brier score	AUROC
5.5	0.458	0.240	0.734
6.0 ^a	0.351	0.181	0.786
6.5	0.260	0.143	0.815

Note. ^a Protocol-defined primary cutoff (≥ 6). CDI denotes community decimal intensity. The sensitivity rows relabel and rescore fixed B2 probabilities; B2 is not refit or recalibrated for the alternate cutoffs. In the construction sample, 0.164 of events lie within 0.5 CDI units of the primary cutoff, and the no-skill Brier score is 0.204. Lower Brier score and higher AUROC indicate better performance.

5.2 Difficulty and endpoint sensitivity

Table 3 and the left panel of Fig. 3 report a descriptive fixed-prediction threshold probe. As the severe cut moves from 5.5 to 6.5, holdout prevalence falls smoothly (0.458, 0.351, 0.260) and the source-only logistic-regression Brier score and AUROC move monotonically (Brier 0.240, 0.181, 0.143; AUROC 0.734, 0.786, 0.815). On the construction corpus the near-threshold fraction ($|\text{maxCDI} - 6| < 0.5$) is 0.164 and the no-skill Brier score is 0.204, bounding the difficulty of the primary endpoint. This probe relabels and rescores probabilities fitted for the primary 6.0 endpoint; it does not refit or recalibrate B2 at 5.5 or 6.5 and therefore does not establish endpoint robustness. The protocol-required aggregation-grid sensitivity was not executed and is disclosed as a deviation (Table 8).

366 5.3 Observed split-performance differences

367 The resource’s headline quantity is the optimism of naive random splitting relative
368 to the sequence-grouped protocol, measured for a fixed leakage diagnostic—a random
369 forest with 300 trees and unrestricted depth—under three fold-assignment schemes
370 on the construction corpus (Table 4, Fig. 4). This diagnostic is distinct from the
371 tuned temporal-holdout baseline B4, whose maximum depth is 3. On this $M \geq 5$ one-
372 row-per-event universe the point estimates are small and slightly reversed: sequence
373 grouping produced a Brier score 0.0059 lower and an AUROC 0.0124 higher than
374 random-event splitting, calculated from unrounded scores. The sequence-grouped split
375 therefore scores marginally *better* than the naive random split, not worse. The frozen
376 internal protocol required uncertainty reporting for this contrast but did not define
377 a paired interval estimator; the executed result contains point estimates only. We
378 disclose this unfulfilled element as a deviation and do not invent a post-hoc inter-
379 val. Accordingly, we report the contrast descriptively, not as evidence of equivalence,
380 a true zero effect, leakage, or a win. Its plausible structural context is that 1,885
381 sequences are largely singleton (1,530 in construction), leaving limited within-sequence
382 near-duplicate structure for a random split to leak. This does not prove a small true
383 effect. With one row per event, event grouping imposes the same grouping constraint
384 as random-event assignment, but its deterministic fold allocation differs and there-
385 fore produces different scores. It is reported as an intermediate control. The protocol
386 remains reusable on datasets with more multi-event sequences. The present result nei-
387 ther demonstrates practically important leakage from random splitting nor establishes
388 its absence.

389 5.4 Baseline performance and calibration

390 Table 5 and Fig. 5 give per-baseline holdout performance ($n = 430$) with sequence-
391 cluster bootstrap 95% intervals. The no-skill B0 has Brier 0.232 ($[0.212, 0.253]$); its

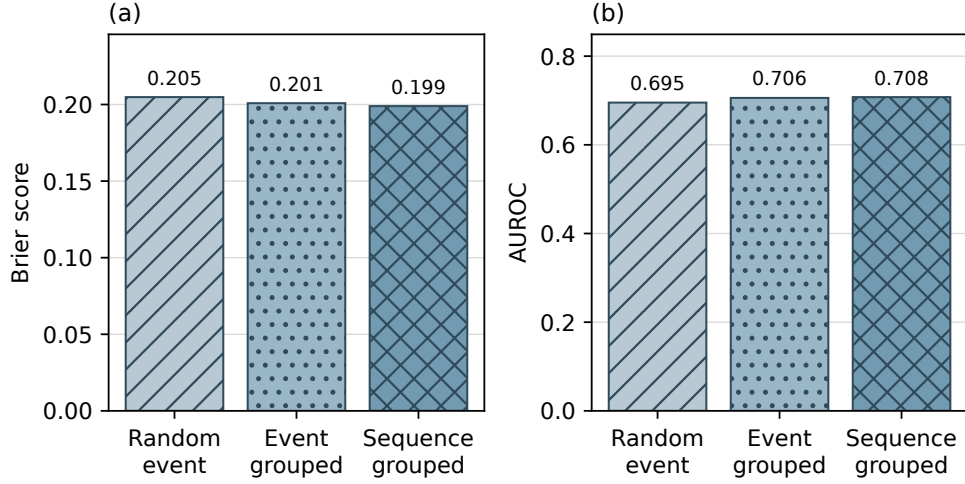


Fig. 4 Split-assignment diagnostic. Five-fold performance of the fixed diagnostic (random forest: 300 trees, unrestricted depth) under random-event, event-grouped, and sequence-grouped fold assignment in the construction sample. Relative to random-event splitting, sequence grouping produced (a) a Brier score 0.0059 lower and (b) an AUROC 0.0124 higher. The diagnostic is distinct from tuned temporal-holdout baseline B4. The small, reversed differences provide no evidence of performance inflation from random splitting in this sample. Both axes start at zero

Table 4 Split-assignment performance of the fixed diagnostic

Fold assignment	Brier score	AUROC	Log-loss	Folds
Random-event (naive)	0.2048	0.6952	0.9640	5
Event-grouped (control)	0.2009	0.7058	0.9223	5
Sequence-grouped (primary)	0.1990	0.7076	0.9709	5

Note. The diagnostic is distinct from the tuned temporal-holdout baseline B4 (depth 3). Using unrounded values, sequence grouping produced a Brier score 0.0059 lower and an AUROC 0.0124 higher than random-event splitting. The frozen internal protocol required uncertainty but did not define the paired interval estimator; the executed report retained point estimates only. This deviation is disclosed, and no post-hoc interval or equivalence claim is added. Because there is one row per event, event grouping imposes the same grouping constraint as random-event assignment, but its deterministic fold allocation is different and can produce different scores.

392 calibration slope cannot be estimated because it assigns the same probability to every
393 event. The learned baselines have lower Brier (B1 0.198, B2 0.181, B4 0.187, B5 0.181)
394 with overlapping intervals; calibration intercept/slope pairs are finite and positive. No
395 baseline is claimed to win; the table reports proper scores, uncertainty, and derived
396 calibration states only.

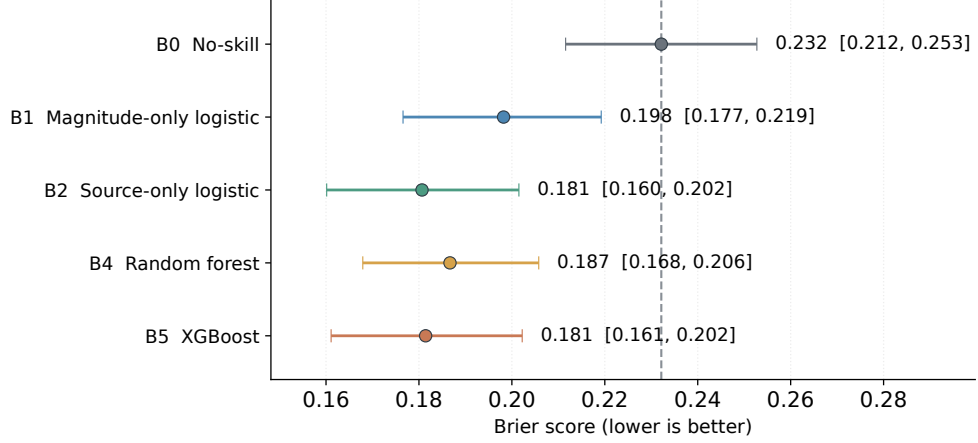


Fig. 5 Temporal-holdout baseline performance. Brier scores and 95% sequence-cluster bootstrap confidence intervals for the five baselines on the temporal holdout ($n = 430$). Lower values are better; the vertical dashed line marks the no-skill B0 score. B3 is omitted because event year is outside the declared source-only temporal-holdout contract (Table 8)

Table 5 Temporal-holdout performance of fixed baseline models

Baseline	Brier	95% CI	Log-loss	AUROC	Calibration ^a	Accuracy
B0 (no-skill)	0.232	[0.212, 0.253]	0.658	0.500	INFEASIBLE	0.65
B1 (magnitude-only logistic)	0.198	[0.177, 0.219]	0.583	0.719	0.32 / 1.02	0.70
B2 (source-only logistic)	0.181	[0.160, 0.202]	0.536	0.786	0.64 / 1.29	0.73
B4 (random forest)	0.187	[0.168, 0.206]	0.553	0.777	0.74 / 1.48	0.73
B5 (XGBoost)	0.181	[0.161, 0.202]	0.541	0.783	0.42 / 1.07	0.74

Note. ^a Intercept / slope. These values cannot be estimated for B0 because it assigns the same probability to every event. B3 includes event year and is therefore omitted from the temporal holdout (Table 8).

Accuracy uses a fixed probability cutoff of 0.5 that was not selected on the holdout. Log-loss clips probabilities to $[10^{-15}, 1 - 10^{-15}]$. Lower Brier score and log-loss are better; higher AUROC and accuracy are better. The two lowest-Brier models cannot be distinguished in their direct paired comparison.

5.5 Worked candidate qualification

Table 6 applies the released engineering decision rule to the six fixed candidates. B3 is outside the source-only contract because it contains event year; this is a conservative scope exclusion, not demonstrated label leakage. B1, B2, B4, and B5 pass the audit and each has a paired Brier interval below the no-skill B0 reference. The contract therefore supports their improvement over B0. It does not support naming B2 or B5 as the winner: their direct paired interval includes zero. This worked example demonstrates the benchmark’s intended use—admit only valid candidates, support improvements with uncertainty, and leave an unresolved comparison unresolved.

Table 6 Worked candidate-model qualification using the frozen temporal holdout

Candidate	Audit gate	Holdout [95% CI]	Brier	Brier difference vs B0 [95% CI]	Contract decision
B0 no-skill	Pass (reference)	0.232 0.253]	[0.212,	—	Reference only
B1 magnitude-only logistic	Pass	0.198 0.219]	[0.177,	-0.0339 [-0.0511, -0.0171]	Supported vs B0
B2 source-only logistic	Pass	0.181 0.202]	[0.160,	-0.0515 [-0.0681, -0.0342]	Supported vs B0
B4 random forest	Pass	0.187 0.206]	[0.168,	-0.0455 [-0.0598, -0.0307]	Supported vs B0
B5 XGBoost	Pass	0.181 0.202]	[0.161,	-0.0507 [-0.0682, -0.0319]	Supported vs B0
B3 expanded-metadata logistic	Out of contract	—	—	—	Year excluded by source-only schema

Decision rule. A candidate is first admitted by the schema, proxy, role, and temporal audits. A lower paired Brier interval that excludes zero supports improvement over the declared reference; otherwise the comparison remains unresolved. All four learned baselines improve over B0. B2 versus B5 remains unresolved (B2–B5 -0.0008, 95% CI [-0.0078, 0.0068]), so the contract prevents a false winner declaration.

5.6 Pairwise ranking stability

Table 7 reports all ten pairwise Brier-score comparisons. Every learned baseline performs better than the no-skill reference B0 (all four intervals exclude zero). Among the learned models, the B2–B4 and B2–B5 comparisons have intervals that include zero. The remaining learned-model intervals exclude zero, although the B4–B5 lower endpoint (0.0003) is near the Monte Carlo boundary. In the joint bootstrap ordering, B2 has the highest probability of being uniquely best (0.613), followed by B5 (0.387), but the direct B2–B5 comparison remains uncertain. These results therefore do not justify naming a winning model.

The frozen internal protocol allowed an “equivalent” verdict but did not specify a positive numerical equivalence margin; its resolution floor was zero. We therefore do not invent a post-hoc margin or label any models equivalent. Instead, we report each point difference, 95% interval, and whether that interval includes zero. Table 8 records this interpretive restriction, which changes no result.

Table 7 Paired Brier-score comparisons among admissible baselines

Model A	Model B	A – B	95% CI	$P(A \text{ better})$	CI relation
B0	B1	0.0339	[0.0171, 0.0511]	0.000	excludes 0
B0	B2	0.0515	[0.0342, 0.0681]	0.000	excludes 0
B0	B4	0.0455	[0.0307, 0.0598]	0.000	excludes 0
B0	B5	0.0507	[0.0319, 0.0682]	0.000	excludes 0
B1	B2	0.0176	[0.0101, 0.0249]	0.000	excludes 0
B1	B4	0.0115	[0.0023, 0.0205]	0.009	excludes 0
B1	B5	0.0168	[0.0055, 0.0273]	0.001	excludes 0
B2	B4	-0.0060	[-0.0118, 0.0002]	0.970	includes 0
B2	B5	-0.0008	[-0.0078, 0.0068]	0.613	includes 0
B4	B5	0.0052	[0.0003, 0.0099]	0.017	excludes 0

Note. Lower Brier score is better, so a negative A – B difference favors model A. Intervals use 2,000 Monte Carlo bootstrap draws; near-zero endpoints are interpreted cautiously and not as stable ranking guarantees. Because no numerical equivalence margin was specified before analysis, intervals including zero are not labelled equivalent. B0 is the no-skill reference; B1 is magnitude-only logistic regression; B2 is source-only logistic regression; B4 is random forest; and B5 is XGBoost.

5.7 Temporal and geographic subgroup summaries

Table 9 and the right panel of Fig. 3 report the temporal and geographic subgroup summaries for the source-only logistic regression within the temporal holdout. The two holdout years are internally consistent (2023: $n = 250$, Brier 0.185, AUROC 0.797; 2024: $n = 180$, Brier 0.174, AUROC 0.777). The geographic subgroup summary spans 82 occupied 10° cells (per-cell Brier 0.002–0.825, median 0.151). These values stratify the same 430 temporal-holdout predictions; they are not a leave-region-out evaluation and support no geographic or cross-system generalization claim.

5.8 Exclusions, deviations, and reproducibility

Table 8 explains the protocol decisions most relevant to interpretation: excluded comparators and records, the unexecuted geographic transport branch, the unfulfilled split-contrast interval, and the absence of an equivalence claim. The tagged public release contains the complete SHA-256 fingerprints, software environment, and machine-readable verification details. Its public-release manifest fingerprints every distributed file. Three original execution-bookkeeping records are self-referential—they cannot contain their own final fingerprints—and are therefore reported separately

Table 8 Protocol exclusions, deviations, and interpretation boundaries

Decision	Treatment	Rationale
B3 expanded-metadata model	Omitted from temporal holdout	Includes event year, which reveals the temporal evaluation period.
Published DYFI intensity equation	Omitted as comparator	Needs unavailable site distance, tectonic-region inputs, and probability conversion.
Two prelisted quarantine IDs	No effect in version 1	Both are absent from the frozen snapshot; no released record is excluded.
Geographic transport evaluation	Not executed	Only temporal-holdout geographic subgroups are reported; no leave-region-out or generalization claim.
Aggregation-grid sensitivity	Not executed	Required but absent; only threshold sensitivity of fixed predictions is reported.
Uncertainty coverage	Partial deviation	The protocol requested intervals for every metric and the split contrast; only Brier-based intervals were retained.
Equivalence interpretation	Descriptive only	No numerical margin was specified; intervals are reported without equivalence labels.

Note. These decisions were recorded before, or disclosed with, the analysis. No result, subset, cutoff, or comparison was omitted because of its observed outcome.

Table 9 Temporal and geographic summaries within the temporal holdout

Subset	Events	Brier score	AUROC
Temporal 2023	250	0.185	0.797
Temporal 2024	180	0.174	0.777
Geographic cells within temporal holdout (82)	430	0.151 (med.)	–

Note. The geographic row partitions the same 430 temporal-holdout predictions into 82 occupied 10° cells. It is a descriptive subgroup summary, not a leave-region-out or external geographic evaluation. The entry is the median cell-level Brier score; cell values range from 0.002 to 0.825 and are included in the public release.

rather than treated as analysis-file mismatches. A machine-readable corrections overlay deprecates three legacy semantic fields (protected-role status, the generic split-interpretation template, and event-group wording) and provides authoritative replacements without changing any numerical result.

440 6 Discussion

441 *Engineering use and decision consequence.*

442 The resource supports a transparent retrospective audit of an event-level classifier.
443 The user does not choose a convenient split or reconstruct a moving dataset; the
444 declared candidate is evaluated against a released contract. The audit rejects forbid-
445 den outcome-derived inputs, the role manifest prevents events linked by the declared
446 heuristic from crossing roles, and the paired Brier interval determines whether improve-
447 ment over a reference is supported. The worked example changes two decisions: B3 is
448 outside the source-only contract, while B2 and B5 are both supported over no-skill
449 but cannot be ranked decisively. Without the contract, the same point scores invite
450 an unsupported winner claim. Because all labels are public, a developer can tune to
451 this release; prospective or confirmatory qualification requires a fresh hidden holdout
452 or independent evaluation service.

453 *Interpreting the split contrast.*

454 The observed point estimate is small and slightly reversed. In these folds the naive
455 random split did not inflate the diagnostic score, but the missing interval pre-
456 vents an equivalence, zero-effect, or “little practical effect” inference. The protocol
457 remains useful for datasets containing more multi-event sequences; this result neither
458 demonstrates leakage nor supports a win.

459 *No winning model.*

460 Every learned baseline separates from the no-skill reference. However, the direct B2–
461 B5 comparison between the two lowest-Brier models is unresolved, as is B2–B4, so
462 the results do not establish a decisive winner. Consistent with the resource’s scope,
463 we claim no winning model and no operational, real-time, lead-time, cross-system, or
464 causal benefit.

465 7 Limitations and threats to validity

466 *Outcome and catalogue limitations.*

467 The endpoint is magnitude-dominated, CDI remains sensitive to responder coverage,
468 and magnitude and depth are frozen preferred catalogue values rather than guaranteed
469 first-alert values. The ten-response rule sets an eligibility and reliability floor but
470 selects on post-event participation and does not remove ascertainment; response count
471 is used only for eligibility. B1 provides the simple magnitude comparator.

472 *Unexecuted sensitivity and transport branches.*

473 The geographic values are temporal-holdout subgroups, not external transport evi-
474 dence. The protocol-defined geographic evaluation and aggregation-grid sensitivity
475 were not executed, and no alternative to the fixed 100-km/30-day sequence defini-
476 tion was tested. The rule guarantees co-assignment only for events it links; it does
477 not guarantee identification of every physical aftershock relationship. Consequently,
478 no geographic, aggregation, sequence-definition, or complete aftershock-isolation
479 robustness claim is made.

480 *Inference and held-out semantics.*

481 The protocol-required split-contrast uncertainty interval was not retained, so that
482 result remains descriptive with no equivalence or zero-effect inference. Non-Brier
483 scores remain descriptive point estimates. Pairwise intervals use 2,000 Monte Carlo
484 draws and near-zero endpoints are interpreted cautiously. The temporal holdout was
485 excluded from model development, but aggregate feasibility was previously known; it
486 is neither “untouched” nor confirmatory.

487 *Public release.*

488 Version 1 of the data, code, frozen results, and reproduction instructions
489 is available in a tagged public repository at <https://github.com/akssha74/>

490 [dyfi-benchmark-reliability](#). The release retains complete digital fingerprints so that
491 later revisions remain distinguishable from the version evaluated here.

492 8 Conclusion

493 Reliable engineering model selection requires more than reporting the lowest point
494 score. The released DYFI benchmark makes a candidate pass an auditable schema
495 and role contract, evaluates it on a fixed temporal holdout, and attaches paired uncer-
496 tainty to the comparison decision. In the worked qualification, four learned candidates
497 improve over no-skill, B3 is outside the declared source-only contract, and the two
498 lowest-Brier models remain unresolved; the contract therefore prevents an out-of-scope
499 comparison and a false winner declaration. The random-versus-sequence diagnostic is
500 small and reversed in this sparse-sequence corpus, so it supplies no evidence of random-
501 split optimism. All results reconstruct from the frozen release. The contribution is a
502 reusable retrospective test harness for trustworthy model comparison. Confirmatory
503 use requires a fresh hidden holdout or independent evaluation service; the release is
504 not a real-time warning, causal model, or operational decision system.

505 Declarations

- 506 • **Funding.** This research did not receive any specific grant from funding agencies in
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- 508 • **Competing interests.** The authors have no relevant financial or non-financial
509 interests to disclose.
- 510 • **Ethics approval.** Not applicable. The study analyzes public, aggregate event-level
511 USGS products rather than individual questionnaire responses; it involved no inter-
512 action with participants, identifiable data, human-subject collection, animals, or
513 biological material.
- 514 • **Consent to participate.** Not applicable; no participants were recruited.

- 515 • **Consent for publication.** Not applicable.
- 516 • **Availability of data and materials.** Version 1 data, results, manifests,
517 and instructions are available under CC0 1.0 at [https://github.com/akssha74/](https://github.com/akssha74/dyfi-benchmark-reliability/releases/tag/v1.1.1)
518 [dyfi-benchmark-reliability/releases/tag/v1.1.1](https://github.com/akssha74/dyfi-benchmark-reliability/releases/tag/v1.1.1). Source data are USGS DYFI and
519 ANSS ComCat (public domain; [10.5066/F7J101C8](https://doi.org/10.5066/F7J101C8) and [10.5066/F7MS3QZH](https://doi.org/10.5066/F7MS3QZH)) and
520 are reconstructable via the released fetch-and-verify source-locator manifest. No
521 respondent-level personally identifying information is redistributed.
- 522 • **Availability of code.** Project: DYFI Benchmark Reliability. Homepage: [https://](https://github.com/akssha74/dyfi-benchmark-reliability)
523 github.com/akssha74/dyfi-benchmark-reliability. Archived version: release v1.1.1.
524 Operating systems: platform independent. Programming language: Python. Licence:
525 MIT. Restrictions on non-academic use: none. Full prediction reconstruction and
526 model tests use the separately pinned scientific-Python environment.
- 527 • **Author contributions.** A.S. conceived and designed the benchmark protocol,
528 developed the software, curated the data, performed the analyses, validated the
529 results, and drafted the manuscript. L.P. contributed to study design and inter-
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